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XTRA LONG

$$v(x_k - x_0) = v(x_k) + v(x_0) - 2\rho^* \sqrt{v(x_k)\, v(x_0)}$$

$$v(x_k) = \sigma^2$$

$$v(x_0) = v\left(\frac{x_1 + \dots x_k}{k}\right) = \frac{1}{k^2}\left[k\sigma^2 + 2\sum_{i<j} \operatorname{cov}(x_i, x_k)\right]$$

$$= \frac{1}{k^2}\left[k\sigma^2 + 2\frac{k(k-1)}{2}\rho\sigma^2\right]$$

EQUAL VARIANCE

$$= \frac{1}{k^2}\left[k\sigma^2 + k(k-1)\rho\sigma^2\right]$$

SAME FOR

$$= \frac{1}{k}\left[\sigma^2 + (k-1)\rho\sigma^2\right]$$

$$= \frac{\sigma^2}{k}\left[1 + (k-1)\rho\right]$$

$$v(x_k - x_0)$$

$$= \sigma^2 + \frac{\sigma^2}{k}\left[1 + (k-1)\rho\right] - 2\rho^* \sqrt{\sigma^2 \frac{\sigma^2}{k}(1 + (k-1)\rho)}$$

$$= \sigma^2\left[1 + \frac{1}{k}(1 + (k-1)\rho) - 2\rho^* \sqrt{\frac{1}{k}(1 + (k-1)\rho)}\right]$$

$$= \sigma^2\left[\frac{k+1}{k} + \frac{k-1}{k}\rho - 2\rho^* \sqrt{\frac{1 + (k-1)\rho}{k}}\right]$$

IF $x_k = \frac{1}{g}\sum x_i$

$$\rho^* = \operatorname{cov}(x_k, x_0)$$

$$= \frac{\operatorname{cov}(x_k, \frac{1}{r}\sum x_i)}{\sqrt{x_k}\sqrt{x_0}}$$

$$= \frac{\frac{1}{r}\sum \operatorname{cov}(x_k, x_i)}{\sqrt{\sigma^2 \frac{\sigma^2}{k}(1 + (r-1)\rho)}}$$

$$= \frac{\rho\sigma^2}{\sigma^2\sqrt{\frac{1}{r}(1 + (r-1)\rho)}}$$

$$v(x_k) = \frac{\sigma^2}{g}(1 + (g-1)\rho)$$

$$\frac{\sigma^2}{g}(1 + (g-1)\rho) + \frac{\sigma^2}{k}(1 + (k-1)\rho) - 2\rho\sqrt{\frac{\sigma^2}{g}(1 + (g-1)\rho)\frac{\sigma^2}{k}(1 + (k-1)\rho)}$$

VAR $(x_k - x_0) = \sigma^2\left[\frac{1}{g}(1 + (g-1)\rho) + \frac{1}{k}(1 + (k-1)\rho) - 2\rho\sqrt{\frac{1}{gk}(1 + (g-1)\rho)(1 + (k-1)\rho)}\right]$?

WHAT IF $\rho_t = \rho_1 (.9)^{t-1}$

GEOMETRIC DECAY

$$\left[\begin{aligned} \rho_2 &= \rho_1 \cdot .9^1 \\ \rho_3 &= \rho_1 (.9)^2 \end{aligned}\right]$$

$$\rho_1 = .85(.9)^{t-1}$$

$$\rho_2 = .85(c)^1$$

$$\rho_4 = .85(c)^3 = .65$$

$$\Rightarrow \sqrt[3]{\frac{.65}{.85}} = c$$

IF

$g = n$

$$\sigma^2\left[\frac{2}{g}(1+(g-1)\rho) - 2\rho\sqrt{\frac{1}{g^2}(1-(g-1)\rho)^2}\right.$$

$$\sigma^2\left[\frac{2}{g}(1+(g-1)\rho) - 2\rho\frac{(1-(g-1)\rho)}{g}\right]$$

$$= \sigma^2\frac{2}{g}(1+(g-1)\rho) - \frac{2\rho}{g}(1-(g-1)\rho)$$

$$= \sigma^2\frac{2}{g}(1+(g-1)\rho)(1-\rho)$$

$$\mathrm{var}(x_k - x_0) = \sigma^2 + \sigma^2 - 2\rho\sigma^2 = 2\sigma^2(1-\rho)$$

TRY V^{-1} w/o a

$$V = UGU' + R$$
$$= U\sigma_1^2 U' + \sigma^2 I$$
$$\sigma^2 V = \frac{\sigma_1^2}{\sigma^2} UU' + I$$
$$\rho = \frac{\sigma_1^2}{\sigma^2} \qquad = \rho UU' + I$$
$$A = I$$

$$(\sigma^2 V)^{-1} = I - IU\left(\frac{1}{\rho} + U'IU\right)^{-1} U'I$$
$$= I - U\left(\frac{1}{\rho} + U'U\right)^{-1} U'$$

$U = t$

$$= I - t\left(\frac{1}{\rho} + 1\right)^{-1} t'$$
$$= I - t\left(\frac{1+\rho}{\rho}\right)^{-1} t'$$
$$\frac{1}{a} V^{-1} = I - \frac{\rho}{1+\rho} tt'$$
$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{\rho}{1+\rho} \begin{pmatrix} t_1^2 & t_1 t_2 \\ t_1 t_2 & t_2^2 \end{pmatrix}$$
$$= 1 - \frac{\rho}{1+\rho} t_1^2$$
$$= 1 - \frac{\frac{b}{a}}{1+\frac{b}{a}} t_1^2 = \left[1 - \frac{b}{a+b} t_1^2\right]$$

$a + b t_2^2$

$a + b(1 - t_1^2)$

$$\frac{a + b - b t_1^2}{a+b}$$

$a + b - b(1 - t_2^2)$

$a + b t_2^2$

OK!

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$X'V^{-1}X$

$$V^{-1} = \frac{1}{a}\left(I - \frac{b}{a+b}tt'\right)$$

$$\frac{1}{a}\begin{pmatrix} 1 & 0 & -\frac{b}{a+b}t_1^2 & -\frac{b}{a+b}t_1t_2 \\ 0 & 1 & -\frac{b}{a+b}t_1t_2 & -\frac{b}{a+b}t_2^2 \end{pmatrix}$$

$$X'\frac{1}{a}\begin{pmatrix} 1-\frac{b}{a+b}t_1^2 & -\frac{b}{a+b}t_1t_2 \\ \frac{-b}{a+b}t_1t_2 & 1-\frac{b}{a+b}t_2^2 \end{pmatrix}X$$

$$\frac{1}{a(a+b)}X'\begin{pmatrix} a+b-bt_1^2 & -bt_1t_2 \\ -bt_1t_2 & a+b-bt_2^2 \end{pmatrix}X$$

$$\frac{b}{a(a+b)}X'\begin{pmatrix} \frac{a+b}{b}-t_1^2 & -t_1t_2 \\ -t_1t_2 & \frac{a+b}{b}-t_2^2 \end{pmatrix}X$$

$$\frac{b}{a(a+b)}\begin{pmatrix} 1 & 1 \\ t & t \\ t & t \end{pmatrix}\begin{pmatrix} & \\ & \end{pmatrix}\begin{pmatrix} 1 & t & t \end{pmatrix}$$

$$\begin{pmatrix} 1 & t & t \\ 1 & t & t \end{pmatrix}^{2\times 3} \qquad \frac{a+b}{b}-t_1^2 \quad -t_1t_2 \quad \left(\frac{a+b}{b}-t_1^2\right)t_1$$

$$\frac{b}{a+b} = \rho \Rightarrow \frac{\rho}{a}X'\begin{pmatrix} \frac{1}{\rho}-t_1^2 & -t_1t_2 \\ -t_1t_2 & \frac{1}{\rho}-t_2^2 \end{pmatrix}X$$

$$\frac{\rho}{a}X'\begin{pmatrix} \frac{1-\rho t_1^2}{\rho} & -t_1t_2 \\ -t_1t_2 & \frac{1-\rho t_2^2}{\rho} \end{pmatrix}$$

$$\frac{1}{a}X'\begin{pmatrix} 1-\rho t_1^2 & -\rho t_1t_2 \\ -\rho t_1t_2 & 1-\rho t_2^2 \end{pmatrix}X$$

2×3

$$\begin{pmatrix} 1-\rho t_1^2-\rho t_1 t_2 & 1-\rho t_2^2-\rho t_1 t_2 \\ (1-\rho t_1^2)t_1-\rho t_1 t_2 t_1 & -\rho t_1 t_2 t_1+(1-\rho t_2^2)t_1 \\ (1-\rho t_1^2)t_1-\rho t_1 t_2 t_1 & -\rho t_1^2 t_2+(1-\rho t_2^2)t_1 \end{pmatrix} \begin{pmatrix} 1 & t & t \end{pmatrix}$$

$3\times 2 \quad 2\times 2$

$$(1,1)\begin{cases} 1-\rho t_1^2-\rho t_2^2-\rho t_1 t_2-\rho t_1 t_2 \\ 1-\rho-2\rho t_1 t_2 \end{cases}$$

$$(2,1)\ t_1(1-\rho t_1^2+1-\rho t_2^2)-2\rho t_2 t_1^2$$
$$t_1(2-\rho\quad)-2\rho t_2 t_1^2$$

$$V^{-1}=\frac{1}{a}\left(I-\frac{b}{a+b}tt'\right)$$

$\rho=\frac{b}{a+b}$

$$\frac{1}{a}(I-\rho tt')$$

$$X'V^{-1}X=\frac{1}{a}(X'X-\rho X'tt'X)$$

a^2+ab $\qquad \rho=\frac{b}{a+b}$

$\cancel{a^2+a(r(a+b))}$

a^2+

$a(a+b)$

$a(b/r)=\frac{ab}{r}$ $\qquad \frac{ab}{\frac{b}{a+b}}$

$a^2+ab=\cancel{a(a+b)}=a(b/r)$

ab/r

$ar(a+b)$

$$\frac{n(a+bt_2^2)}{b} \qquad \frac{n(a+b(1-t_1^2)}{b}$$

$$= \frac{n(a+b-bt_1^2)}{b} = n\left(\frac{a+b}{b} - \frac{bt_1^2}{b}\right)$$

$$= 1 - \cancel{t_1^2}$$

$$= 1 - rt_1^2$$